Exercise 3: Stored Procedures

CODE:

```
CREATE TABLE Accounts (
   AccountID   NUMBER PRIMARY KEY,
   CustomerID  NUMBER,
   AccountType VARCHAR2(20),
   Balance     NUMBER
);

INSERT INTO Accounts VALUES (101, 1, 'Savings', 10000);
INSERT INTO Accounts VALUES (102, 2, 'Savings',  5000);
INSERT INTO Accounts VALUES (103, 3, 'Current',  8000);

COMMIT;

CREATE TABLE Employees (
   EmployeeID   NUMBER PRIMARY KEY,
   EmployeeName VARCHAR2(50),
   Department   VARCHAR2(30),
   Salary       NUMBER(10, 2)
);

INSERT INTO Employees VALUES (1, 'Rahul', 'IT',      50000);
INSERT INTO Employees VALUES (2, 'Priya', 'HR',      45000);
INSERT INTO Employees VALUES (3, 'Rohan', 'IT',      60000);
INSERT INTO Employees VALUES (4, 'Anita', 'Finance', 55000);
```

```sql
COMMIT;

SELECT * FROM Accounts;
SELECT * FROM Employees;

-- Scenario 1: Apply 1% monthly interest to all Savings accounts
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
AS
BEGIN
   UPDATE Accounts
   SET Balance = Balance + (Balance * 0.01)
   WHERE AccountType = 'Savings';

   COMMIT;
END;
/

EXEC ProcessMonthlyInterest;

SELECT * FROM Accounts;

-- Scenario 2: Update employee bonus by department and percentage
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(
   dept         IN VARCHAR2,
   bonusPercent IN NUMBER
```

```
)
AS
BEGIN
   UPDATE Employees
   SET Salary = Salary + (Salary * bonusPercent / 100)
   WHERE Department = dept;

   COMMIT;
END;
/

-- Scenario 3: Transfer funds between accounts
EXEC UpdateEmployeeBonus('IT', 10);

SELECT * FROM Employees;

CREATE OR REPLACE PROCEDURE TransferFunds(
   sourceAcc IN NUMBER,
   destAcc   IN NUMBER,
   amount    IN NUMBER
)
AS
   balance NUMBER;
BEGIN
   SELECT Balance
   INTO balance
```

```
    FROM Accounts
    WHERE AccountID = sourceAcc;

    IF balance >= amount THEN
        UPDATE Accounts
        SET Balance = Balance - amount
        WHERE AccountID = sourceAcc;

        UPDATE Accounts
        SET Balance = Balance + amount
        WHERE AccountID = destAcc;

        COMMIT;
        DBMS_OUTPUT.PUT_LINE('Transfer Successful');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Insufficient Balance');
    END IF;
END;
/
```

OUTPUT:

All rows fetched: 3 in 0.174 seconds

	ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE
1	101	1	Savings	10000
2	102	2	Savings	5000
3	103	3	Current	8000

17 CREATE TABLE Employees (

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULT SCRIPT O

All rows fetched: 4 in 0.160 seconds

	ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE
1	101	1	Savings	10100
2	102	2	Savings	5050
3	103	3	Current	8000
4	104	4	Savings	20200

All rows fetched: 4 in 0.152 seconds

	EMPLOYEEID	EMPLOYEENAME	DEPARTMENT	SALARY
1	1	Rahul	IT	60500
2	2	Priya	HR	45000
3	3	Rohan	IT	72600
4	4	Anita	Finance	55000

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULT SCR

All rows fetched: 4 in 0.160 seconds

	ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE
1	101	1	Savings	6100
2	102	2	Savings	9050
3	103	3	Current	8000
4	104	4	Savings	20200